MTH309 PRACTICE PROBLEMS WEEK 12

Question 1. Let $\{X_n\}$ and $\{Y_n\}$ be two sequences of random variables on the same probability space. If $\sum_{n=1}^{\infty} \mathbb{P}(X_n \neq Y_n) < \infty$, then show that $\sum_n (X_n - Y_n)$ converges a.s.. If $\{a_n\}$ be a sequences of positive real numbers such that $a_n \uparrow \infty$, then show that $\frac{1}{a_n}\sum_n (X_n - Y_n)$ converges to 0 a.s..

Question 2. For a sequence $\{X_n\}$ of random variables on the same probability space, show the following:

(i) If $X_n \xrightarrow[n\to\infty]{a.s.} 0$, then $\frac{X_1+X_2+\cdots+X_n}{n} \xrightarrow[n\to\infty]{a.s.} 0$.

(ii) If $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p} 0$, then $\frac{X_1+X_2+\cdots+X_n}{n} \xrightarrow[n\to\infty]{\mathcal{L}^p} 0$, for any fixed $p, 1 \leq p < \infty$.

(iii) If $X_n \xrightarrow[n\to\infty]{\text{in prob.}} 0$, then show by an example that $\frac{X_1+X_2+\cdots+X_n}{n} \xrightarrow[n\to\infty]{\text{in prob.}} 0$ need not be true.

Question 3. Let $U^{(n)} := \{x = (x_1, \cdots, x_n)^t \in \mathbb{R}^n : \sum_{j=1}^n x_j^2 < 1\}$. Let $\alpha_n := \lambda^{(n)}(U^{(n)})$, where $\lambda^{(n)}$ denotes the Lebesgue measure on $\mathbb{R}^n$. Show that $\alpha_n = \frac{2\pi}{n}\alpha_{n-2}$ for all integers $n \geq 3$.

Question 4. Suppose $X_n \xrightarrow[n\to\infty]{\mathcal{L}^1(\mathbb{P})} X$ and $Y_n \xrightarrow[n\to\infty]{\mathcal{L}^\infty(\mathbb{P})} Y$. Show that $X_nY_n \xrightarrow[n\to\infty]{\mathcal{L}^1(\mathbb{P})} XY$.

Question 5. Let $\{X_n\}_n$ be a sequence of independent RVs with $S_n := X_1 + \cdots + X_n, \forall n \geq 1$. Assume that (Etemadi's inequality)

$$\mathbb{P}(\max_{1\leq j\leq n} |S_j| \geq 3\epsilon) \leq \max_{1\leq j\leq n} \mathbb{P}(|S_j| \geq 3\epsilon).$$

Show that $\{S_n\}_n$ converges in probability if and only if $\{S_n\}_n$ converges a.s..

Question 6. Let $\{X_n\}_n$ be a sequence of iid $Poisson(4)$ RVs. Let $\bar{X}_n := \frac{1}{n}(X_1 + \cdots + X_n), \forall n \geq 1$. Find the limit $\lim_{n\to\infty} \sin(\pi\bar{X}_n)$ in the sense of convergence in probability. Does the same limit exist in the sense of a.s. convergence?